Pre-gravity6plane: the Pin(4,4) fermion field dirac16complex, its 6-planes, gravity, Kohn-Sham pair creation, Einstein-Rosen bridges and deflating degrees of freedom

This repository extends the Mathematica notebook Pre-gravity6plane.nb (P. L. Nash, (C) 2022, GNU General Public License) with a second-quantized fermion field, dirac16complex, and the calculations requested for it. Every equation and result stated below is computed with exact arithmetic (or a convergent numerical method where stated) in the notebooks 6planes.nb (Sections 1 to 4) and deflation.nb (Section 5).

1. Setting

Coordinates $x_0, \ldots, x_7$ (x_0 hidden space, $x_{1,2,3}$ space, x_4 time, $x_{5,6,7}$ superluminal time), metric $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$. The eight real 16×16 Dirac matrices are those of Pre-gravity6plane.nb,

$$T^A = (T16^A)[A] = \begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}, \qquad T^A T^B + T^B T^A = 2\eta^{AB} I_{16}, \qquad S^{AB} = \tfrac{1}{4}[T^A, T^B],$$

with entries in $\{-1,0,1\}$; the S^{AB} generate Spin(4,4) and, with the reflections T^C, Pin(4,4); $\Gamma = T^0\cdots T^7 = \mathrm{diag}(-I_8, I_8)$, $P_{L,R} = \frac{1}{2}(I_{16} \mp \Gamma)$, $\sigma_{16} = T^0T^1T^2T^3$.

2. The field and its Lagrangian

dirac16complex $\psi = (\psi_0, \ldots, \psi_{15})$ has complex anticommuting (Grassmann) components and is a second-quantized fermion field; $\bar{\psi} = \psi^\dagger \sigma_{16}$.

$$\mathcal{L} = \tfrac{1}{2}\left(\bar{\psi} T^A \partial_A \psi - (\partial_A \bar{\psi}) T^A \psi\right) - m\bar{\psi}\psi - \frac{\lambda}{2}(\bar{\psi}\psi)^2, \qquad T^A\partial_A\psi = (m + \lambda\bar{\psi}\psi)\psi.$$

$\mathcal{L}$ is exactly real; the quadratic (mass) and quartic (self-coupling) terms are non-zero because the components are complex. Quantization with time x_4 gives $\{\psi_a, \psi_b^\dagger\} = K_{ab}\delta$, $K = -i\sigma_{16}T^4$, $K^2 = I$, with an (8,8) indefinite (Krein) metric, and an exact split into two mirror sectors of opposite energy: $\mathcal{L} = \mathcal{L}_D[\psi_+] - \mathcal{L}_D[\psi_-] - \frac{\lambda}{2}(S_+ - S_-)^2$ for fields independent of x_5, x_6, x_7.

3. Mirror of Pre-gravity6plane.nb and the 6-planes

Restricted to configurations that depend only on x_0 and x_4, the analogue of the notebook's $\Psi_{16}(x_0, x_4)$ ($m = -2M$), the 16 field equations split into the four blocks $\{0,5,8,13\}, \{1,4,9,12\}, \{2,7,10,15\}, \{3,6,11,14\}$, as in the old notebook; its exponential solutions $e^{pz+qt}u$ with $(6p+q)(6p-q) = m^2/H^2$, $z = 6Hx_0$, $t = Hx_4$, are exactly the Maple solutions of the old notebook. With the triality frames and vector forms $V_1 = \eta(\psi_1^T \sigma F_A{}^a)$, $V_2 = F_a{}^A\psi_2$, a Universe defines the 6-plane

$$M^6 = \{Z \in \mathbb{R}^8 : n_1 \cdot Z = 0,\ n_2 \cdot Z = 0\}, \qquad n_k = V_k[\mathrm{Re}\,\psi],$$

with $\binom{8}{2} = 28$ parameterizations; a pair of universes of energies $\pm E$ creates the 6-plane $\{Z : \mathrm{Re}V_2 \cdot Z = \mathrm{Im}V_2 \cdot Z = 0\}$ by itself.

4. Gravity, Kohn-Sham DFT, pair creation, Einstein-Rosen bridges

Three gravitational fields (diagonal vielbeins): primordial $\mathrm{diag}(\tan 6Hx_0,\ e^{-Hx_4}\sin^{-1/6} 6Hx_0\,(\times 3),\ 1,\ e^{Hx_4}\sin^{-1/6} 6Hx_0\,(\times 3))$ (the vielbein of Pre-gravity6plane.nb with $a_4 = Hx_4$); modern $\mathrm{diag}(f^{-1/2}, 1, 1, 1, f^{1/2}, 1, 1, 1)$ with $f = 1 - r_s/x_0$; future the same with $f = 1 - r_s/x_0 - H_\Lambda^2 x_0^2$. In each, the Euler-Lagrange equations derived from the Grassmann Lagrangian are $\gamma^\mu D_\mu\psi = (m + \lambda\bar{\psi}\psi)\psi$, e.g. (primordial)

$$\cot(6Hx_0)T^0\partial_0\psi + e^{Hx_4}\sin^{\frac{1}{6}}(6Hx_0)\sum_{i=1}^{3}T^i\partial_i\psi + T^4\partial_4\psi$$
$$+ e^{-Hx_4}\sin^{\frac{1}{6}}(6Hx_0)\sum_{j=5}^{7}T^j\partial_j\psi - 3H\cot^2(6Hx_0)T^0\psi = (m+\lambda\bar{\psi}\psi)\psi.$$

The quartic term is treated in a Kohn-Sham (DFT) approximation, $M = m+\lambda\rho_s$, with the Kohn-Sham Hamiltonian $h_{KS} = \alpha p + BM\beta$ ($\alpha = -T^4T^0$, $\beta = -iT^4$) in tortoise coordinates, ground and first-excited (ΔSCF) states, and the finite-temperature thermodynamic effective potential $\Omega(T)$. Proven in each field: $\mathcal{L}_m[\Gamma\psi^*] = \mathcal{L}_{-m}[\psi]$ and $\mathcal{L}_m[T^1\psi(R_1x)] = \mathcal{L}_{-m}[\psi]$ (mass-sign pairing); $T^0h_{KS}T^0 = -h_{KS}$ and $h_{KS}^* = -h_{KS}$ (every level ϵ has a partner $-\epsilon$); in the zero-energy vacuum the creation of a + universe pair together with a − universe pair changes the energy by exactly 0, while single quanta and single-sign creation are forbidden by the conserved charges and energy: universes of masses $\{+m, -m\}$ are created in pairs. The 1935 Einstein-Rosen bridges are verified; the modern field has one and the future field two exact bridges (two congruent sheets, $g = 0$ on the bridge, finite curvature); the primordial field's degenerate hypersurface $x_0 = 0$ is a curvature singularity ($R \approx -\frac{31}{216H^2x_0^4}$); across a bridge the vacuum pairs universes of energy $+\epsilon$ and $-\epsilon$ with weight $1/(e^{\epsilon/T_H}+1)$, $T_H = \kappa/2\pi$.

5. Deflating degrees of freedom

In deflation.nb the superluminal times x_5, x_6, x_7 of the primordial field are replaced by deflating degrees of freedom ($z = 6Hx_0$):

$$ds^2 = \tan^2 z\,dx_0^2 + e^{-2Hx_4}\sin^{-1/3} z\,(dx_1^2+dx_2^2+dx_3^2) - dx_4^2 - g_{55}\,dx_5^2 - g_{66}\,dx_6^2 - g_{77}\,dx_7^2,$$
$$g_{jj} = b_{jj}e^{-ax_4}, \quad a > 0,\ b_{jj} > 0.$$

Proper sizes along x_5, x_6, x_7 shrink as $\sqrt{b_{jj}}\,e^{-ax_4/2}$. The Ricci scalar, the Einstein tensor and the Kretschmann scalar depend on x_0 only, so deflation creates no curvature singularity at finite x_4. The pure deflation field $dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2 - dx_4^2 - \sum_j g_{jj}dx_j^2$ is flat $\mathbb{R}^4$ times a hyperbolic 4-space of radius $2/a$ whose four directions are all timelike,

$$-dx_4^2 - \sum_{j=5}^{7} g_{jj}\,dx_j^2 = -\frac{4}{a^2}\,\frac{dw^2+dy_5^2+dy_6^2+dy_7^2}{w^2}, \qquad w = \frac{2}{a}\,e^{ax_4/2}, \quad y_j = \sqrt{b_{jj}}\,x_j,$$

with $R = 3a^2$ and Kretschmann scalar $3a^4/2$. World lines with momentum $p_5 = -g_{55}\,dx_5/d\ell$ (ℓ an affine parameter) turn back in time at $x_4^* = \frac{1}{a}\ln(b_{55}/p_5^2)$. The field equation of dirac16complex, derived exactly from the Grassmann Lagrangian, is

$$\cot z\,T^0\partial_0\psi + e^{Hx_4}\sin^{\frac{1}{6}} z\sum_{i=1}^{3}T^i\partial_i\psi + T^4\partial_4\psi + e^{ax_4/2}\sum_{j=5}^{7}\frac{T^j\partial_j\psi}{\sqrt{b_{jj}}}$$
$$-\frac{3}{2}H\cot^2 z\,T^0\psi - \frac{3}{2}\left(H+\frac{a}{2}\right)T^4\psi = (m+\lambda\bar{\psi}\psi)\psi.$$

Its universes $e^{i(k_0x_0+k_1x_1+k_2x_2+k_3x_3+q_5x_5+q_6x_6+q_7x_7-\omega x_4)}u$ (frozen coefficients) obey

$$\omega^2 = m^2 + \cot^2 z\,k_0^2 + e^{2Hx_4}\sin^{1/3} z\,(k_1^2+k_2^2+k_3^2) - e^{ax_4}\sum_{j=5}^{7}\frac{q_j^2}{b_{jj}},$$

so momentum along a deflating direction turns oscillation into exponential growth, and the critical deflation rate is $a_c = 2H$. Every mode equation $\chi' = d\chi/dx_4 = M\chi$ with $M = T^4(i\sum_{A\neq4}c_AT^A - m)$ and real c_A satisfies $M^\dagger K + KM = 0$. So the Krein charge $\chi^\dagger K\chi = P_+ - P_-$ is conserved, where $P_\pm = \chi^\dagger\frac{1}{2}(I_{16}\pm K)\chi$ are the weights of the +mass and -mass sectors, while only the deflating directions change the norm $P_+ + P_-$. Hence P_+ and P_- grow by equal amounts: deflation creates {+mass, -mass} universes in pairs. In the exact numerical example the -mass fraction $P_-/(P_+ + P_-)$ grows from 1.2×10^{-31} to 0.49999995.

6. Assumptions and interpretations made in this work

1. The modern and future gravitational fields (not specified in the request) are defined as above; the primordial field is the notebook's vielbein with $a_4(Hx_4) = Hx_4$.
2. "Universe" means a mode function (c-number solution) of dirac16complex, as the notebook calls Ψ_{16} the wave function of a Universe; its "mass" is its energy for translations of x_4 (or the mass parameter m for the mass-sign theorem).
3. The theorems are exact statements about this model; they are not a proof about the physical universe.
4. In Section 5 the deflating components enter the line element as $-g_{jj}dx_j^2$ and all other components are those of the primordial field, as the user specified; the dispersion relation there uses frozen (adiabatic) coefficients.

7. Reproducing everything

```
cd code
wolframscript -file verify_dirac_matrices.wls ../Pre-gravity6plane.nb
wolframscript -file build_nb2.wls
wolframscript -file build_deflation.wls
python3 gen_dm.py
python3 md2tex.py <file>.md <file>.tex ; pdflatex <file>.tex
```

8. Repository inventory

file	content
Pre-gravity6plane.nb	original notebook (P. L. Nash, GPL)
6planes.nb	new notebook, evaluated, 34 input cells, no messages
deflation.nb	new notebook on the deflating degrees of freedom, evaluated, 17 input cells, no messages
README.md/.tex/.pdf	this overview
dirac matrices.md/.tex/.pdf	proofs and display of the eight real 16x16 Dirac matrices
dirac16complex_6planes.md/.tex/.pdf	the field, Lagrangian, quantization, mirror of the old notebook, 6-planes
gravity_DFT_pair_creation.md/.tex/.pdf	three gravitational fields, Euler-Lagrange equations, Kohn-Sham DFT, pair theorems
einstein_rosen_bridges.md/.tex/.pdf	Einstein-Rosen bridges
deflation.md/.tex/.pdf	deflating degrees of freedom: geometry, curvature, geodesics, field equation, universes, pair creation
*_provenance.md/.tex/.pdf	sources and computations behind each document
*_symbols.md/.tex/.pdf	definitions of all symbols of each document
code/	core.wl, verify_dirac_matrices.wls (+ log), dm_data.json, gen_dm.py, md2tex.py, 6planes_src.txt, build_nb2.wls (+ log), deflation_head.txt, deflation_tail.txt, deflation_src.txt, build_deflation.wls (+ log)